VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Mission System

AFLCMC/WA

April 24, 2026

U.S. GOVERNMENT NOTICES: Furnished to the U.S. Government with UNLIMITED RIGHTS as defined in DFARS 252.227-7013, this takes precedence over any restrictive marking herein.

Table of Contents

1 Overview **1**
1.1 Purpose . . . 1
1.2 Scope . . . 1
1.3 Authoritative References . . . 1
1.4 Informative References . . . 2

2 Mission System **3**
2.1 High Level Architecture . . . 4
2.2 Minimum Procurable Units . . . 4
2.2.1 MPU Types . . . 6
2.2.2 MPU Fault Reporting . . . 7
2.3 Compute Cluster Farm . . . 9
2.4 Special Signals . . . 10
2.4.1 Special Signals Category Allowability . . . 10
2.4.2 Specific Special Signals Restrictions . . . 11
2.5 Interaction with non-AMS GRA Elements . . . 15

3 Compliance Requirements **16**

A Acronyms **23**

B Revision History **25**

List of Figures

2.0.1 AMS GRA Mission System Components 3
2.2.1 MPU Classes and Their Relationships 5
2.3.1 CCF Decomposition 9

List of Tables

1.3.1 Authoritative References 2
1.4.1 Informative References 2

2.2.1 MPU Software Conditions 8
2.4.1 Special Signals Category Table 11
2.4.2 GRA Tailored OMS Special Signals 14

3.0.1 Mission System Requirements 22

B.0.1 Revision History Table 25

1.0 Overview

An Agile Mission Suite (AMS) Government Reference Architecture (GRA)-compliant Mission System is a physical embodiment of legacy Mission System elements, AMS GRA-defined elements, and interfaces to non-Mission System platform subsystems.

1.1 Purpose

The Mission System Volume describes both the physical and logical aspects of an AMS GRA-compliant Mission System. Furthermore, this Volume provides details on architectural considerations likely needed by a Mission System but not wholly specified by AMS GRA. The purpose of the Mission System Volume is to provide guidance on how the physical and logical elements of a Mission System can integrate into, and are supported by, the architecture. The Volume also provides compliance requirements against which an AMS GRA-compliant Mission System is assessed.

1.2 Scope

The Mission System Volume is applicable to entities seeking AMS GRA compliance or performing a compliance assessment of a Mission System.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or Minimum Procurable Unit (MPU) provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the Mission System Volume.

Standard	Title	Version/Date
AMS GRA Volumes	MASI Volume	14.0
AMS GRA Volumes	Armament Volume	14.0
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Time Distribution Volume	14.0
AMS GRA Volumes	AMS GRA Service MPU Volume	14.0
AMS GRA Volumes	CDS MPU Volume	14.0

Standard	Title	Version/Date
AMS GRA Volumes	Countermeasures Manager (CMM) MPU Volume	14.0
AMS GRA Volumes	Cryptographic Unit MPU Volume	14.0
AMS GRA Volumes	DPP MPU Volume	14.0
AMS GRA Volumes	Data Storage MPU Volume	14.0
AMS GRA Volumes	Local Function MPU Volume	14.0
AMS GRA Volumes	MFA MPU Volume	14.0
AMS GRA Volumes	OMS Service MPU Volume	14.0
AMS GRA Volumes	PCaB MPU Volume	14.0
AMS GRA Volumes	Switch MPU Volume	14.0
AMS GRA Models	AMS GRA Architecture Model	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the Mission System.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Mission System Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Software Architecture Volume	14.0
AMS GRA Volumes	Security Framework Volume	14.0
AMS GRA Volumes	Resource Management Volume	14.0
AMS GRA Volumes	Pilot Vehicle Interface Volume	14.0
AMS GRA Volumes	Executive Summary	14.0
AMS GRA Volumes	Cyber IDD	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Mission System and Related Entities.

2.0 Mission System

Figure 2.0.1 describes the composition of an AMS GRA Mission System.

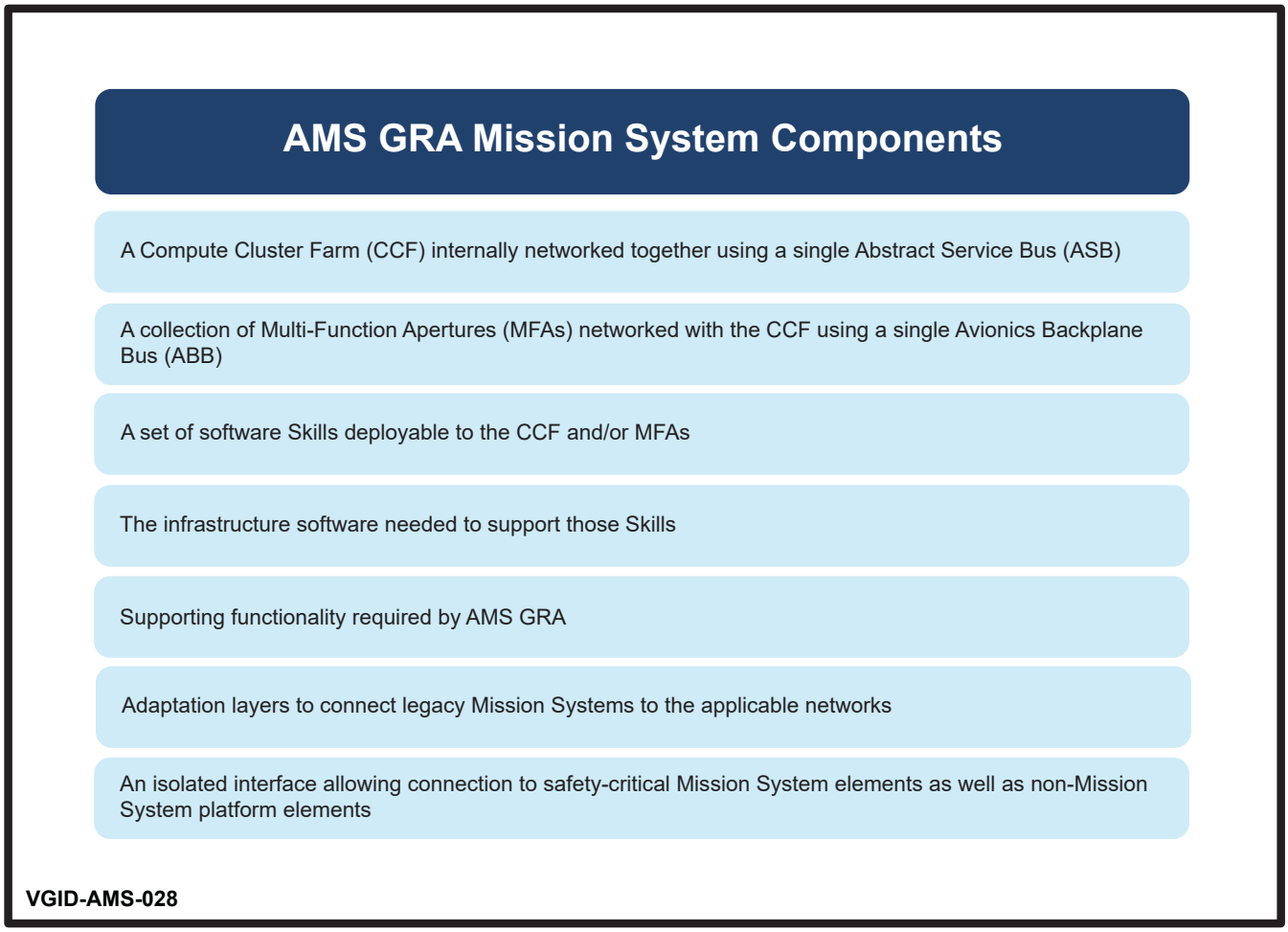


Figure 2.0.1: AMS GRA Mission System Components Enable a Mission System to Be Compliant with AMS GRA Requirements.

AMS GRA does not define the physical characteristics of any individual MPU, the number or configuration of the collection of MPUs, or any specific network topology for the required networks. It is the responsibility of the MSI and Procuring Authority to coordinate with MPU developers to ensure the physical elements of AMS GRA integrate together to fulfill the mission requirements. The MSI is given the freedom and the responsibility to define the necessary networks to meet mission needs.

The Mission System is also responsible for distributing time, defining the overall security framework, and providing isolation between AMS GRA and safety-critical elements of the platform and the Pilot Vehicle Interface (PVI). Details can be found in the respective Volumes.

2.1 High Level Architecture

Logically, the Mission System is divided into three primary layers with additional supporting functionality: the Multi-Function Aperture (MFA) layer, the Multi-Function Processor (MFP) layer, and the Mission Processing Subsystem (MPS) layer. The Avionics Backplane Bus (ABB) supports communication between the MFA layer and the MFP layer as well as within the MFP layer. The Abstract Service Bus (ASB) supports communication between the MFP layer and the MPS layer as well as within both the MFP and MPS layers.

See the Executive Summary for various representations of how the MFA, MFP, and MPS layers interact. One representation of this interaction is depicted in the AMS GRA Architecture on a Page graphic within the Executive Summary.

See the Software Architecture Volume for further detail on how these interactions between the three primary layers fit into the context of a Mission System.

2.2 Minimum Procurable Units

The foundation of AMS GRA is the MPU. MPUs are categorized into Levels 1 through 3. The choice of MPU level is driven by procurement time horizons, technology evolution cycles, and platform integration limitations. **Figure 2.2.1** illustrates the set of AMS GRA MPUs and the levels these MPUs belong to. Further detail describing the three levels can be found in **Section 2.2.1** of this Volume, as well as in the Executive Summary.

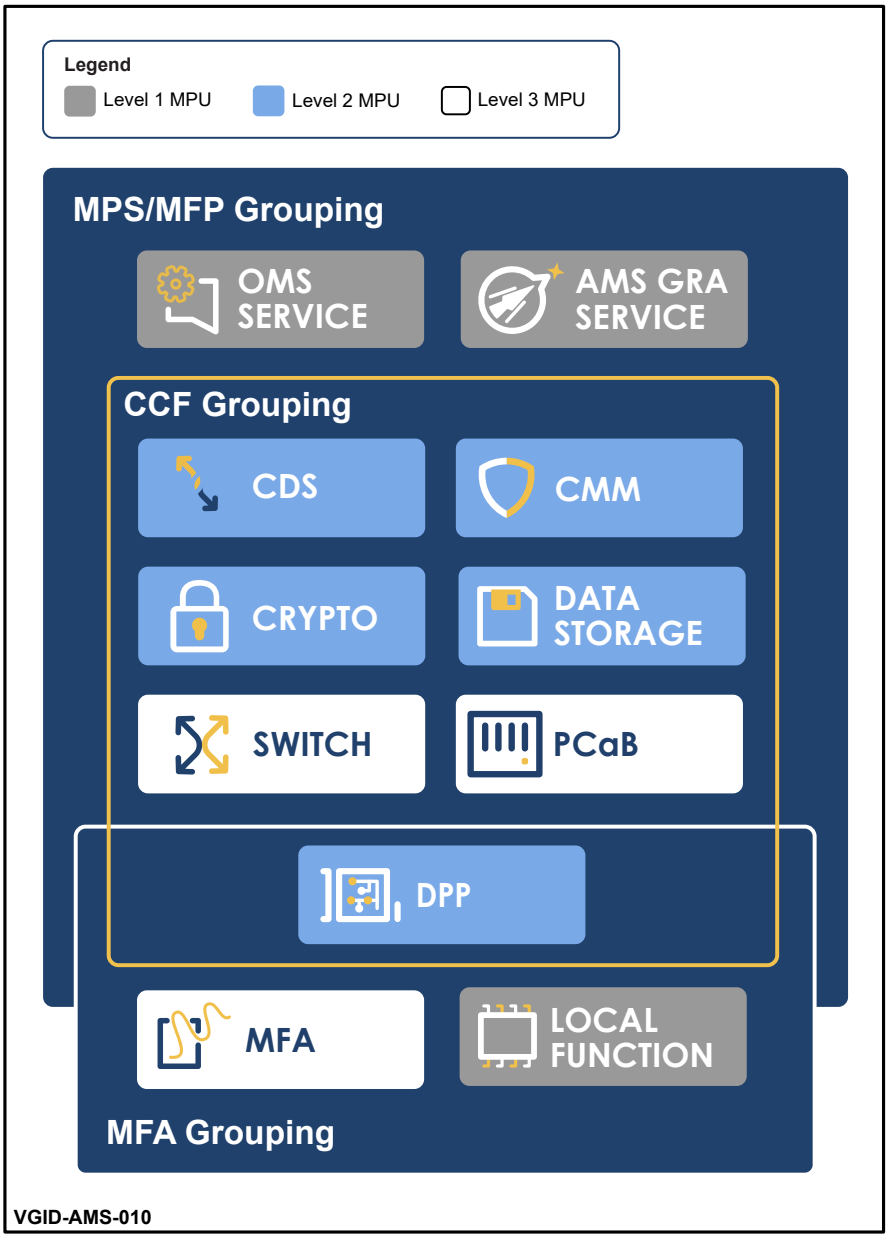


Figure 2.2.1: MPU Levels Help Define the Role of MPUs and Frame Requirements for AMS GRA Compliance.

MPU Compliance

MPU levels play an important role in deriving the requirements of each MPU. Requirements prescribe relevant methods of verification to meet the various MPU compliance tiers. The Compliance Volume discusses the concept of MPU compliance tiers in more detail. For MPUs to be AMS GRA compliant, they must satisfy compliance requirements specific to the MPU and MPU compliance tier an MSI wants to achieve.

MPU Volumes

Each MPU volume contains a more complete definition of key features, context, interfaces, and compliance requirements, as follows:

- Key features provide design guidance on what is typically expected for the MPU type[1].
- Context diagrams contain a graphical representation of the allowed, conditionally required, and required interfaces from the perspective of the MPU.
- Interface tables contain a textual representation of the interfaces and their descriptions.
- Compliance Requirements describe the components an MPU must provide to achieve a given compliance tier.

2.2.1 MPU Types

The platform provider is responsible for the initial selection of Level 3 MPUs on any given platform in order to facilitate platform installation details.

The Level 3 MPUs establish a grouping of MPUs that mature at rates slower than the Level 1 MPU and, in many cases, the Level 2 MPUs. Although many of these MPUs constituent technologies may advance at steady rates driven by commercial industry, the ability to refresh these technologies is limited due to heavy platform or airframe design integration. The significant time and cost required to insert new Level 3 MPUs is what separates them into their own class.

The set of Level 3 MPUs consists of both Radio Frequency (RF) and Infrared (IR) MFAs as well as elements that reside within the MFP and MPS logical constructs. The MFAs are an integral part of the thrust to achieve total spectrum dominance in a manner consistent with AMS GRA. In general, all MFAs provide a digital data stream requiring processing prior to consumption by Mission System elements. A by-product of the convergence of RF and IR MFAs is a common lexicon. AMS GRA acknowledges this and provides unique terms, where appropriate, to describe differences between aperture spectral types.

[1] Reference MFA type definitions provided in the Glossary.

The Level 2 MPUs establish a grouping of capabilities that evolve at rates slower than the Level 1 MPU due to the maturation rate of the underlying technologies. Many of the Level 2 MPUs are either entirely driven by commercial industry or leverage components that are commercially driven. The development and integration timeframe of these MPUs is expected to be shorter than Level 3 MPUs.

In AMS GRA, there are three types of Level 1 MPUs: AMS GRA Services, Open Mission Systems (OMS) Services, and MFA Local Functions. Services enable third-party capability insertion into MPS and MFP layers of the AMS GRA software architecture. MFA Local Functions similarly enable third-party capability insertion into the MFA layer of the architecture.

Services combine to implement the behaviors of an AMS GRA Mission System at run time. These behaviors include not only mission-specific software logic such as Electronic Support Measures (ESM), external communications, radar, and Infrared Search and Track (IRST), but also mission-agnostic logic that is likely to be present in any Mission System. AMS GRA does not prescribe the presence of specific Services, nor does it prescribe how a Mission System partitions the behaviors of a software Skill into Services. At run time, the collection of Services implements an architectural design pattern in which Services run independently of one another and communicate between themselves through the passing of messages. This architectural pattern helps to enforce loose coupling between Services, an important enabler of several business and technical objectives.

2.2.2 MPU Fault Reporting

In order to improve Mission System integration and maintainability, AMS GRA standardizes fault reporting definition across MPUs. Consistent fault reporting definition allows MPU users (e.g. monitoring services) to better determine a system anomaly's source and root cause once that anomaly is detected by the AMS GRA. The AMS GRA fault reporting definition supports fault reporting at the MPU's component level which can pertain, but is not limited to, the following:

- Line Replaceable Unit (LRU)
- MPU components that the MSI and/or Government end user services or maintains
- MPU components that impact known capability

When an unexpected event occurs in an MPU, the event can be categorized as one of the following:

- Recoverable with no effect on responsibilities: The event has no impact on the MPU after recovery.
- Recoverable with effect on responsibilities: The event has limited impact on the MPU after recovery and that MPU is still able to meet its responsibilities or reduced responsibilities.
- Non-recoverable: The event stops the MPU from meeting its responsibilities.

However, it is important to note that some of the tasks required to generate the fault report may not be possible depending on the type, severity, timing, and location of the fault (e.g., power failure). Regardless, the MPU attempts to complete all applicable tasks.

MPU Fault Reporting Approach

MPUs may report faults for any of the event categories listed above. Fault reports provided to the Mission System include fault status containing the following information: unique identification of the fault occurrence, fault code, and fault severity. Unique identification of the fault occurrence is traceable to the MPU, preferably to the MPU's component. The fault severity for an MPU is aligned with the values of the Universal Command and Control Interface (UCI)-defined fault severity enumeration type and should be either Nominal, Caution, Warning, or Failed. The fault code is aligned with the UCI-defined fault code type. Level 1 MPUs have additional conditions, as identified in **Table 2.2.1**.

Category	Additional Level 1 MPU Conditions
Recoverable with no effect on responsibilities	The Service or Local Function is uniquely identifiable. If applicable, a message with a log severity level of Info is added to the service log [2] .
Recoverable with effect on responsibilities	The Service or Local Function is uniquely identifiable. If applicable, a message with a log severity level of Warning or Error is added to the service log [2].
Non-recoverable	The Service or Local Function is uniquely identifiable. If applicable, a message with a log severity level of Fatal is added to the service log [2]. If applicable, a Service exception occurs.

Table 2.2.1: MPU Software Conditions Specify the Additional Conditions Levied on Level 1 MPUs When an Unexpected Event Occurs for a Given Category.

The MSI ensures all reported faults are aggregated such that fault reports generated at any layer (MFA/MFP/MPS) are available through OMS messaging. AMS GRA does not mandate that all MPUs have access to the ASB, but does mandate that MPU fault reports have a path to the ASB (e.g., monitoring or translation Service).

AMS GRA considers UCI messages to be the preferred method for reporting faults. This methodology is reflected in the tiering structure associated with fault reporting compliance requirements. Commercial-Off-The-Shelf (COTS)

[2]The inclusion of service log support depends on the Level 1 MPU designs. The service log fault severity is aligned to the UCI defined log severity enumeration type and is an indication of the severity level of the logged message.

products and other Mission System components may not natively communicate over a software interface or in UCI format; however, they can still be used as MPU components. For example:

- The MPU provider includes software capability that translates analog or discrete signals from the component to a UCI or Non-UCI software interface.
- The MPU provider includes a software library that translates Non-UCI information to a UCI format. [3] .

2.3 Compute Cluster Farm

The Compute Cluster Farm Compute Cluster Farm (CCF) comprises a variable number of Processor Chassis and Backplane Processor Chassis and Backplane (PCaB) MPUs, Data Storage MPUs, Switch MPUs, Digital Payload Processor Digital Payload Processor (DPP) MPUs, Countermeasures Manager CMM MPUs, Cryptographic Unit MPUs, and Cross Domain Solution Cross Domain Solution (CDS) MPUs as necessary to complete overall platform objectives. The physical configuration of the CCF takes into account actual network latency, bandwidth, and redundancy requirements in determining the system configuration rather than logical system boundaries, such as Multi-Function Processor MFP and Mission Processing Subsystem MPS.

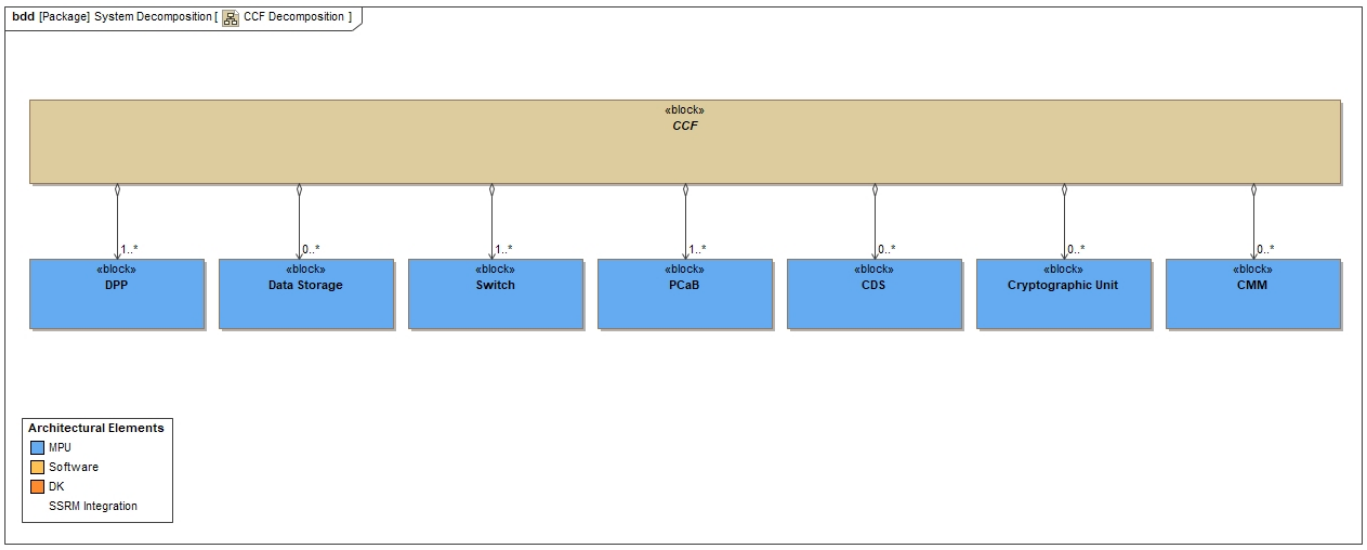


Figure 2.3.1: The Logical Decomposition of the CCF may contain up to seven distinct MPUs wherin at least one DPP MPU, Switch MPU, and a PCaB MPU are required.

[3]The inclusion of service log support depends on the Level 1 MPU designs. The service log fault severity is aligned to the UCI defined log severity enumeration type and is an indication of the severity level of the logged message.

2.4 Special Signals

OMS Special Signals represent a class of interfaces that are normative, non-message-based signals that typically leverage a hardware-based or analog-based solution, such as discretes. These OMS signals are used where latency, safety, or security concerns preclude the use of a message-based communication construct. The OMS Standard defines a specific and limited set of valid Special Signals for an OMS instantiation. OMS currently organizes this set of Special Signals into seven categories:

1. Discretes
2. Time Reference Signals
3. Frequency Reference/Intermediate Frequency
4. High-Speed Data Lines
5. Other
6. Excluded
7. Superseded Signals

AMS GRA allows for the utilization of the OMS Special Signal definitions under certain conditions. Furthermore, AMS GRA leverages these definitions to provide compatibility with existing OMS compliant Mission System elements. However, in order to meet the goals of portability, upgradability, and commonality, AMS GRA further constrains the set of Special Signals defined by OMS for an AMS GRA compliant MPU. These added constraints reflect the applicability and necessity of specific OMS Special Signals in order to support development of AMS GRA MPUs.

2.4.1 Special Signals Category Allowability

Table 2.4.1 captures the seven categories of OMS Special Signals and defines their allowability and AMS GRA usage (if allowed).

Name	Documentation	OMS Tailoring - GRA Allowed	OMS Tailoring - GRA Openness Tailoring
Discretes	Specific discretes are allowed and disallowed, as described in **Table 2.4.2**.	Allowed	Common

Name	Documentation	OMS Tailoring - GRA Allowed	OMS Tailoring - GRA Openness Tailoring
Time Reference Signals	Allowed for AMS GRA defined MPUs that include a Platform Time interface as an allowed interface. See specific MPU Volume for further definition of allowed interfaces.	Allowed	Common
Frequency Reference/Intermediate Frequency	Within AMS GRA, this interface is explicitly allowed only from the Platform to the MPU. This interface is not allowed as external MPU interface to another MPU. Since AMS GRA does not control internal interfaces of MPUs, this special signal may be allowable in specific implementations within the MPU.	Allowed	Open Platform Defined
High-Speed Data Lines	Specific High-Speed Data Lines are allowable as described in **Table 2.4.2**.	Allowed	Common
Other	Allowed for MPUs that have Key Fill interface defined	Allowed	Common
Excluded	Not allowed for MPU external interfaces	Disallowed	Common
Superseded	Not allowed for MPU external interfaces	Disallowed	Common

Table 2.4.1: Special Signals of Specific Types Are Either Allowed or Disallowed for AMS GRA with Openness Tailoring Values of Common, Platform-Defined, or Open.

2.4.2 Specific Special Signals Restrictions

Table 2.4.2 describes the restrictions on specific OMS Special Signals with respect to AMS GRA usage. Each allowed signal is specifically denoted as either Platform Defined (Open), Common, or Disallowed. The table documents the rationale for continued support of the signal.

Name	Documentation	OMS Tailoring - OMS Special Signals Category	OMS Tailoring - GRA Openness Tailoring	OMS Tailoring - GRA Allowed
Subsystem Present	This signal is supported by use cases for minimizing Mission System startup time (to skip time allocated to a system if not present) and for integrated test with changing configurations.	Discretes	Platform Defined Open	Allowed
Subsystem On	This signal is supported by use cases for minimizing Mission System startup time and sequencing initialization to prevent brown-out.	Discretes	Platform Defined Open	Allowed
Emergency Mode	This signal is supported by use cases for Emergency Mode or Combat Override, and as a secondary path for FSDA mission critical functions if applicable.	Discretes	Platform Defined Open	Allowed
Safety Interlock	This signal is supported by use cases for safety concerns.	Discretes	Common	Allowed
Data Load	This signal is used to satisfy safety and security where two independent inputs are required to enter Data Load mode.	Discretes	Common	Allowed
Zeroize/Erase	This signal is supported by use cases for a Security Critical Discrete that needs to propagate to the Cryptographic Units and CDSs and any DPP that involves the movement, usage, or storage of CPI, key materials, or the like.	Discretes	Common	Allowed
Blanking Pulse	This signal is supported by use cases for low-latency coupled pulsed transmit capabilities which induces interoperability concerns with receiving systems needing to be informed of blanking.	Discretes	Common	Allowed
Transmit Override/Inhibit	This signal requests that a Subsystem transmission be inhibited while the signal is active. This signal is supported by use cases for safety concerns (such as temporary vehicle power bus limitations) and interoperability at timescales unsupported by AMS GRA messaging.	Discretes	Common	Allowed

Name	Documentation	OMS Tailoring - OMS Special Signals Category	OMS Tailoring - GRA Openness Tailoring	OMS Tailoring - GRA Allowed
Receive Override/Inhibit	This signal tells a Subsystem to stop receiving operations. The signal is supported by use cases for interoperability.	Discretes	Platform Defined Open	Allowed
Platform ID	This signal is Disallowed as typical use cases for this in AMS GRA should utilize messaging instead. Specifically, AMS GRA MPUs should not depend on external identification of platform uniqueness other than AMS GRA defined messaging interfaces.	Excluded		Disallowed
Subsystem Configuration	This signal is supported by use cases such as physical configuration, or type/presence of removable modules.	Discretes	Platform Defined Open	Allowed
Rotor Blade Sensor	This signal is supported by use cases for Rotary Wing Mission Platforms where rotor velocity information must be conveyed for responsiveness to air vehicle systems and Mission System.	Discretes	Platform Defined Open	Allowed
Shared Aperture Available	This signal is Disallowed as typical use cases for this in AMS GRA should utilize messaging instead.	Superseded		Disallowed
1PPS	This signal is supported by use cases for OMS Legacy Time interfaces. Specifically, 1PPS use is Allowed at all compliance Tiers.	Time Reference Signals	Common	Allowed
BCD Time	This signal is supported by use cases for OMS Legacy Time interfaces including use specified in ICD-GPS-060A/B and may be used to achieve precision time in some legacy systems. Specifically, BCD Time use Allowed at Tier 1, and Disallowed at Tier 2 and Tier 3.	Time Reference Signals	Common	Allowed

Name	Documentation	OMS Tailoring - OMS Special Signals Category	OMS Tailoring - GRA Openness Tailoring	OMS Tailoring - GRA Allowed
Time Fault	This signal is supported by use cases for OMS Legacy Time interfaces including use specified in ICD-GPS-060A/B and in some legacy systems. Specifically, BCD Time use Allowed at Tier 1, and Disallowed at Tier 2 and Tier 3.	Time Reference Signals	Common	Allowed
Key Fill	This signal is supported by use cases for the current NSA Standard Mechanism for encryption key loading.	Other	Open	Allowed
ABB	This signal is an OMS-aligned description of the AMS GRA required ABB interface, which is logically separated from the OMS ASB. The ABB is a tailored OMS Special Signal High-Speed Data Line with an open interface.	High-Speed Data Lines	Open	Required
DGT	This signal is an OMS-aligned description of the AMS GRA DGT interface.	High-Speed Data Lines	Common	Allowed
Other HSDL	This signal is an OMS-aligned description of any other HSDLs utilized for non-AMS GRA standardized architectural elements in an instantiation of a AMS GRA Mission System, such as helmet displays and inter-DPP communication allowable as intra-MPU interfaces.	High-Speed Data Lines	Platform Defined Open	Special Approval

Table 2.4.2: OMS Special Signals May Be Restricted Based on Supporting Use Cases.

2.5 Interaction with non-AMS GRA Elements

Historically, one of the main challenges of upgrading a Mission System is the interaction of the Mission System elements with both Air Vehicle Safety of Flight and Weapon Carriage subsystems. These respective subsystems undergo time-consuming and expensive certification procedures to ensure platform safety. Fortunately, these subsystems change infrequently over the life of an air vehicle. However, any change that modifies or interacts with them may invoke recertification of safety-related air vehicle subsystems.

The Mission System may leverage adaptation and isolation patterns to ensure that updating one or more MPUs does not trigger air vehicle recertification. An OMS Isolator may be used to perform message format conversion, message protocol conversion, and rule-set application that enables and enforces cross-cutting concerns for isolation between the Mission System and the platform.

In any implementation of AMS GRA, expectations are that some AMS GRA elements will interact with non-AMS GRA elements (e.g., legacy subsystems, future non-AMS GRA systems, and platform support Services). As the interaction of the latter with AMS GRA elements supports mission needs, AMS GRA accommodates the interactions by using adapter(s) and, when necessary, by using the same isolation mechanisms leveraged for flight critical element isolation. The MSI assesses the impact to communication and mission execution timelines due to the physical and logical isolation of non-AMS GRA subsystems—including weapons.

3.0 Compliance Requirements

Table 3.0.1 describes the required Mission System characteristics. See the Mission Agnostic Service Infrastructure (MASI) Volume guidance for designing and implementing the allocated Mission Agnostic Service Infrastructure (MASI) requirements. See Cyber Interface Description Document (IDD) and Security Framework Volumes for the details of the Cyber and CDS elements in support of these Security requirements.

Id	Text	Compliance Tier	Recommended Verification Method
GRA-CYAGE-002	A Cyber Agent shall be able to send and receive the messages defined in the Cyber Agent Messages section of the Cyber ICD.	Tier 1	Demonstration
GRA-CYAGE-003	A Cyber Agent shall execute actions at the direction of the Cyber Manager.	Tier 2	Demonstration
GRA-CYAGE-004	A Cyber Agent shall possess functionality and interfaces to execute actions defined within the Cyber Agent Messages section of the Cyber ICD.	Tier 2	Demonstration
GRA-CYAGE-005	A Cyber Agent shall communicate with a Cyber Manager using secure communication.	Tier 3	Analysis
GRA-CYMAN-002	A Cyber Manager shall be able to send and receive the messages defined in the Cyber Manager Messages of the Cyber ICD.	Tier 1	Demonstration
GRA-CYMAN-003	A Cyber Manager shall generate state, mode, and event detail logs.	Tier 1	Demonstration
GRA-CYMAN-004	A Cyber Manager shall be permitted to reconfigure enforcement policies of the Cyber Monitor at runtime using the Cyber Manager messages defined in the Cyber ICD.	Tier 2	Demonstration
GRA-CYMAN-005	A Cyber Manager shall orchestrate the collection of the data from Cyber Monitors.	Tier 2	Demonstration
GRA-CYMAN-006	A Cyber Manager shall dynamically reconfigure a Cyber Monitor using the Cyber Manager Messages defined in the Cyber ICD.	Tier 2	Demonstration
GRA-CYMAN-007	A Cyber Manager shall detect anomalous behavior indicative of cyber events within the Mission System.	Tier 2	Demonstration
GRA-CYMAN-008	A Cyber Manager, IAW its current policy configuration, shall orchestrate the reactions of a Cyber Agent.	Tier 2	Demonstration
GRA-CYMAN-009	A Cyber Manager shall communicate with a Cyber Monitor using secure communication.	Tier 1	Inspection
GRA-CYMAN-010	A Cyber Manager shall communicate with a Cyber Agent using secure communication.	Tier 3	Inspection
GRA-CYMON-002	A Cyber Monitor shall be able to send and receive the messages defined in the Cyber Monitor Messages section of the Cyber ICD.	Tier 1	Demonstration
GRA-CYMON-003	A Cyber Monitor shall possess functionality and interfaces to retrieve data defined within the Cyber Monitor Messages section of the Cyber ICD from MPUs.	Tier 2	Demonstration
GRA-CYMON-004	A Cyber Monitor shall be reconfigurable at runtime by the Cyber Manager using the Cyber Monitor messages defined in the Cyber ICD.	Tier 3	Demonstration
GRA-CYMON-005	A Cyber Monitor shall communicate with a Cyber Manager using secure communication.	Tier 1	Inspection
GRA-MASI-001	A MASI shall be able to start mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration
GRA-MASI-002	A MASI shall be able to stop mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-003	A MASI shall monitor the Management Interface for reported health and status of AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-006	A MASI shall monitor fault information reported by AMS GRA-compliant MPUs.	Tier 1	Inspection
GRA-MASI-008	A MASI shall consume all available BIT information reported by AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-010	A MASI shall provide a mechanism to aggregate disparate fault, health, and status data.	Tier 2	Inspection
GRA-MASI-011	A MASI shall provide a mechanism to store fault, health, and status data.	Tier 2	Inspection
GRA-MASI-012	A MASI shall be able to respond to fault information reported by AMS GRA-compliant MPUs.	Tier 2	Analysis
GRA-MASI-013	A MASI shall provide an interface to access fault, health, and status data.	Tier 2	Inspection
GRA-MASI-015	A MASI shall monitor the health and status reporting of all deployed AMS GRA and OMS Services.	Tier 1	Inspection
GRA-MASI-016	A MASI shall monitor compute resource information (e.g., utilization, availability, capability, bandwidth, flash, RAM).	Tier 2	Inspection
GRA-MASI-017	A MASI shall provide a mechanism to allow services to access all available software and firmware configuration status and versioning information for software and firmware stored and deployed on the AMS GRA Mission System.	Tier 3	Demonstration
GRA-MASI-020	A MASI shall provide a mechanism to consolidate duplicate fault data.	Tier 2	Inspection
GRA-MASI-021	A MASI shall include an OCI-compliant Orchestration Engine.	Tier 2	Inspection
GRA-MASI-022	A MASI shall deploy OCI-compliant containerized software utilizing a container framework.	Tier 1	Demonstration
GRA-MASI-023	A MASI shall provide access to compute resource information.	Tier 2	Inspection
GRA-MASI-024	A MASI shall facilitate endpoint connections between services.	Tier 1	Demonstration
GRA-MASI-025	A MASI shall provide a mechanism to allow services to access Local Function information.	Tier 2	Inspection
GRA-MASI-027	A MASI shall register the services available for deployment.	Tier 1	
GRA-MASI-028	A MASI shall report the state of MASI	Tier 1	
GRA-MASI-029	A MASI shall support deployment of a minimum of two instances of the same Service.	Tier 1	Test
GRA-MASI-030	A MASI shall isolate AMS GRA and OMS Service software faults from other software MPUs.	Tier 1	Test
GRA-MASI-031	A MASI shall assign an MPUs' deployment targets and resources in accordance with that MPUs's delivered service.yml file.	Tier 1	Test

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-032	A MASI shall provide an implementation of the AMS Common Transport API.	Tier 1	
GRA-MASI-033	A MASI shall report the status of all SW MPUs available in the system in UCI format.	Tier 2	Test
GRA-MASI-034	A MASI shall report compute resource information in UCI format if requested in UCI format by an MPU within the system.	Tier 2	Test
GRA-MS-001	A Mission System shall implement a single physical ABB that supports the AMS GRA MPU interfaces that use the ABB.	Tier 1	Inspection
GRA-MS-002	A Mission System shall implement a single physical ASB IAW OMS standard.	Tier 1	Inspection
GRA-MS-003	A Mission System shall implement network accessible Data Storage.	Tier 1	Inspection
GRA-MS-004	A Mission System shall implement a documented architecture that supports the distribution of time to support platform-implemented MPUs including PTP, DGT, and/or OMS legacy timing.	Tier 1	Inspection
GRA-MS-005	A Mission System shall implement access and information flow isolation between safety critical and non-safety critical processing.	Tier 1	Analysis
GRA-MS-006	A Mission System that includes SMS shall implement the WSI as defined in the Armament Volume.	Tier 1	Inspection
GRA-MS-007	A Mission System shall implement an API on DPPs, as applicable, to allow AMS GRA Services and OMS Services to access OMS Special Signals.	Tier 2	Inspection
GRA-MS-008	A Mission System shall implement an API on DPPs, as applicable, to allow AMS GRA Services and OMS Services to access OMS platform-specific Elements.	Tier 2	Inspection
GRA-MS-009	A Mission System shall implement an OMS Critical Abstraction Layer on DPPs to allow AMS GRA Services and OMS Services to access to UCI messaging.	Tier 1	Inspection
GRA-MS-010	A Mission System shall provide an AMS GRA-compliant MASI implementation.	Tier 1	Inspection
GRA-MS-013	A Mission System that segregates data into multiple security domains shall utilize an approved Transfer CDS to pass data between security domains.	Tier 3	Analysis
GRA-MS-014	A Mission System utilizing an MLS bus shall enforce logical segregation of security domains by utilizing an approved MLS CDS at each bus interface.	Tier 3	Analysis
GRA-MS-020	A Mission System shall preserve availability of a trusted copy of Mission System software, firmware, and configuration files during a failed reprogramming attempt.	Tier 3	Test
GRA-MS-023	A Mission System shall make available all MEL Implementations supplied by MFA providers to any AMS GRA Service to be loaded.	Tier 1	Analysis
GRA-MS-024	A Mission System shall provide connectivity to the ASB from any DPP in the CCF.	Tier 1	Analysis
GRA-MS-025	A Mission System shall provide connectivity to the ABB from any DPP performing MFP layer processing within the CCF.	Tier 1	Analysis

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-026	A Mission System shall make available to MASI the locality between DPPs, and DPPs to MFAs to inform the deployment and recovery strategy.	Tier 2	Inspection
GRA-MS-027	A Mission System shall provide the capability of all AMS GRA Services to be deployed to any ABB-connected compatible DPP.	Tier 1	Analysis
GRA-MS-028	A Mission System employing MLS logical separation on an MLS bus shall provide interoperable authentication and integrity mechanisms consistent for communicating endpoints in an approved manner.	Tier 3	Analysis
GRA-MS-029	The Mission System shall provide open documentation for the incorporated MLS solution regarding programming of security policies and switch configuration if multiple security domains or security policies are present in support of approved certification.	Tier 3	Inspection
GRA-MS-030	A Mission System shall include non-proprietary documentation for interfaces operating at multiple security domains.	Tier 3	Inspection
GRA-MS-031	A Mission System shall include MLS interface documentation with descriptions of boundaries and allowable data flows between multiple interconnected security domains.	Tier 3	Inspection
GRA-MS-032	A Mission System shall include interface documentation for hardware requirements and interface design details when multiple interconnected security domains are present within the Mission System.	Tier 3	Inspection
GRA-MS-033	A Mission System with classified data shall encrypt all classified DaR using approved cryptographic algorithms.	Tier 1	Inspection
GRA-MS-034	A Mission System shall include a model consistent with MBSE content contained in the AMS GRA SK release with which the Mission System is compliant.	Tier 2	Inspection
GRA-MS-035	A Mission System shall use approved cryptographic algorithms to authenticate and verify the integrity of all Mission System data, software and firmware prior to updating, loading, and executing.	Tier 3	Inspection
GRA-MS-036	The Mission System documentation shall document the implementation of the approved cryptographic authentication and verification algorithms for all loadable Mission System data, software and firmware prior to updating, loading, and executing.	Tier 3	Inspection
GRA-MS-037	A Mission System shall generate audit logs for security relevant events to include generation of audit logs, response to audit failure, protection of audit information and enforcement of non repudiation.	Tier 2	Inspection
GRA-MS-041	A Mission System shall provide storage for Local Functions to be accessible to the associated MFA using networked Data Storage, non-volatile MFA local storage, or non-volatile DPP local storage.	Tier 2	Inspection

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-045	A Mission System shall include SAST results for all microprocessor software source code in the Mission System, IAW the Mission System documentation template.	Tier 1	Inspection
GRA-MS-046	A Mission System shall include DAST results for all microprocessor software in the Mission System, IAW the Mission System documentation template.	Tier 1	Inspection
GRA-MS-049	A Mission System with CPI in non-volatile memory shall prevent recovery of CPI and protective keys in the event of a loss of Mission Platform.	Tier 1	Analysis
GRA-MS-050	A Mission System shall provide interface connectivity for Required MPU interfaces.	Tier 1	Inspection
GRA-MS-051	A Mission System shall provide configurability for audit of security relevant events.	Tier 2	Inspection
GRA-MS-052	A Mission System shall securely deploy software containing CPI.	Tier 1	Demonstration
GRA-MS-054	A Mission System shall include documentation on the deployment and recovery strategy for its software elements.	Tier 1	Inspection
GRA-MS-055	A Mission System shall include non-proprietary machine readable documentation for the security audit log format.	Tier 2	Inspection
GRA-MS-056	A Mission System with MFAs shall provide an implementation of the Resource Management Framework for management of sensor resources.	Tier 1	Inspection
GRA-MS-057	The Mission System shall provide a RoT to Mission System MPUs.	Tier 2	Analysis
GRA-MS-059	A Mission System shall verify the integrity of a hardware MPU over a secure communications channel prior to enabling the transfer of software, mission data, or associated Key Material to/from that MPU.	Tier 2	Demonstration
GRA-MS-060	The Mission System shall orchestrate secure communications for ABB and ASB connected endpoints.	Tier 1	Analysis
GRA-MS-061	The Mission System shall authorize access to endpoints on a per-session basis.	Tier 2	Analysis
GRA-MS-062	The Mission System shall establish secure communications between endpoints using NSA-approved cryptographic methods.	Tier 2	Inspection
GRA-MS-063	The Mission System shall implement dynamic authorization for secure communications.	Tier 3	Analysis
GRA-MS-064	The Mission System shall dynamically revoke session authentication when directed by an authorized source.	Tier 3	Analysis
GRA-MS-065	The Mission System shall only allow authorized communication channels IAW a file identified and documented by the Mission System.	Tier 2	Demonstration
GRA-MS-066	The Mission System shall mutually authenticate all endpoints.	Tier 2	Analysis
GRA-MS-067	The Mission System shall provide a KMP.	Tier 2	Inspection

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-068	The Mission System shall protect security functions from unauthorized access or modification.	Tier 1	Demonstration
GRA-MS-069	The Mission System shall protect security relevant data from unauthorized access or modification.	Tier 1	Demonstration
GRA-MS-071	The Mission System shall use secure communications to transfer classified key material between Mission System endpoints.	Tier 1	Demonstration
GRA-MS-072	The Mission System shall use secure communications to transfer classified software between Mission System endpoints.	Tier 1	Demonstration
GRA-MS-073	The Mission System shall use the MEL Implementation supplied by the MFA provider to establish the interface between an AMS GRA Service and the MFA.	Tier 1	Demonstration
GRA-MS-074	A Mission System shall restrict access to OMS Special Signals through DPP APIs to non-message based signals that cannot be feasibly implemented using AMS GRA- defined common or open forms of communication.	Tier 2	Inspection
GRA-MS-075	A Mission System shall provide a Critical Program Information (Critical Program Information (CPI)) analysis, to include Consequence of Compromise (Consequence of Compromise (CofC)), in accordance with the Department of Defense Anti-Tamper Engineering Framework.	Tier 1	Inspection
GRA-MS-076	A Mission System shall instantiate the Cyber Agent.	Tier 3	Demonstration
GRA-MS-077	A Mission System shall instantiate the Cyber Monitor.	Tier 3	Demonstration
GRA-MS-078	A Mission System shall instantiate the Cyber Manager.	Tier 3	Demonstration
GRA-MS-079	A Mission System shall publish the Service health and status messages on the ABB to the ASB	Tier 1	Inspection

Table 3.0.1: A Mission System's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
BCD	Binary Coded Decimal
BIT	Built-In Test
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CMM	Countermeasures Manager
CofC	Consequence of Compromise
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DGT	Distributed Guaranteed Time
DPP	Digital Payload Processor
ESM	Electronic Support Measures
FSDA	Fail Safe Design Analysis
GRA	Government Reference Architecture
HSDL	High-Speed Data Lane
IAW	In Accordance With
ICD	Interface Control Document
IDD	Interface Description Document
IR	Infrared
IRST	Infrared Search and Track
KMP	Key Management Plan
LRU	Line Replaceable Unit

MASI	Mission Agnostic Service Infrastructure
MBSE	Model-Based Systems Engineering
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MLS	Multi-Level Security
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
OCI	Open Container Initiative
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PPS	Pulse Per Second
PTP	Precision Time Protocol
PVI	Pilot Vehicle Interface
RAM	Random Access Memory
RF	Radio Frequency
RoT	Root of Trust
SAST	Static Application Security Testing
SK	Starter Kit
SMS	Stores Management System
UCI	Universal Command and Control Interface
WSI	Weapon Systems Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision adds additional volumes and models to the Authoritative References Table.	July 24, 2025	SK13.0
B	This revision updates authoritative references and requirements prose.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Mission System Volume Between Documentation Releases.